Yongheng Han

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Academic Interests

I am interested in geometry and analysis, especially in minimal surface and mean curvature flow.

Employment

- **Institute of Mathematics, AMSS, CAS**
  Postdoctoral fellow 2026-2028.

Education

University of Science and Technology of China

- Ph.D. of mathematics 2021-2026.
- Advisor: Bing Wang.

Beijing Institute of Technology

- B.S. of mathematics, 2017-2021.
- Advisor: Lining Jiang.

Publications and Accepted Papers

3. **Ancient mean curvature flows from minimal hypersurfaces**. Geom. Dedicata, 220.15, 2026.
2. **Mean curvature flow with pinched curvature integral** Differ. Geom. Appl. 99,2025, pp:102244
1. **Constructions of minimal Hermitian matrices related to a C^*-subalgebra of $M_n(\mathbb{C})$**(joint with Lining Jiang and Ying Zhang). Proc. Am. Math. Soc, 151.1, 2023, pp:73-84.

Preprints

5. **Singularities of mean curvature flow with bounded total curvature**(joint with Zhen Wang), arXiv:26.
4. **Well-posedness of mean curvature flow**, arXiv:2603.22809.
3. **The Covariant Riesz Transforms on Riemannian Manifolds**(joint with Bing Wang), arXiv:2603.22963.
2. **The Caldón-zygmund inequality on evolving manifolds**(joint with Bing Wang), arXiv:2603.22742
1. **Singularities of mean curvature flows with bounded mean curvature and Morse index**, arXiv:2501.05489.

[1]Updated November 4, 2026

Talks

2. *The Riesz transforms and Calderón-Zygmund inequalities on Riemannian manifolds.* Geometry and Analysis Conference, Tsinghua University, Jun 15-18 2026.

1. *Singularity of mean curvature flow with bounded mean curvature and Morse index.* Seminar, Westlake University, Oct 2025.

Teaching Assistant

University of Chinese Academy of Sciences

- 2026 fall B01GB002H-A02 Calculus I-A (Instructed by Youde Wang) UCAS.

University of Science and Technology of China

- 2025 spring 001506 Equations of Mathematical Physics A (Instructed by Yongbing zhang) USTC
- 2024 fall 001505 Complex Analysis A (Instructed by Yongbo Tian) USTC